complaint-router

Lab 3: LLM baseline and retrieval

KSE MLOps · Lab 3 · Zakhar Staryk · github.com/goodfrog33/kse-mlops-complaint-router · tag lab-3

0.601 vs 0.633
macro-F1, LLM vs Lab 2 model

0.50 → 0.725
retrieval hit rate at 5

$0.0002
per regulation answer

At a glance

LLM baseline

- Same task, same 5,212 test complaints
- 0.601 macro-F1 vs 0.633: the model stays

Retrieval

- 6 CFPB regulations, 195 sections, DVC
- pgvector index, one command rebuilds it

Evaluation

- 40 questions, hit rate, MRR, LLM judge
- 3 sweeps: chunk, embedding, top-k

Cost and latency

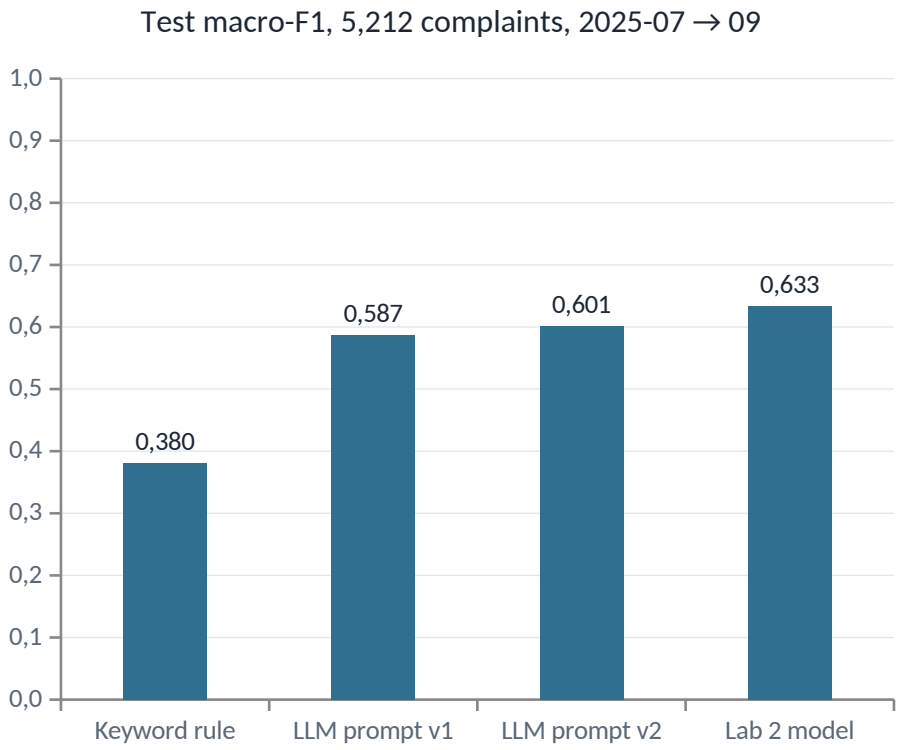
- Tokens, $ and p50/p95 on every run
- API chosen over local vLLM, trade-off stated

The LLM gets two jobs, neither of them routing

	A. LLM baseline	B. Regulation Q&A
Job	Predict the product class, zero-shot	Answer an agent's question from the rules
Input → output	Complaint → one of 10 classes	Question → ≤ 4 sentences + [section]
Corpus	None: the complaint itself	eCFR: Reg B, E, F, V, X, Z (195 sections)
Metric	Test macro-F1, as in Lab 2	Hit rate@k, MRR, judge score
Logged in	MLflow complaint-router	MLflow complaint-router-rag

Model: anthropic/claude-haiku-5.5 via OpenRouter (OpenAI-compatible), temperature 0, reasoning off

Same test set, same metric: the trained model stays



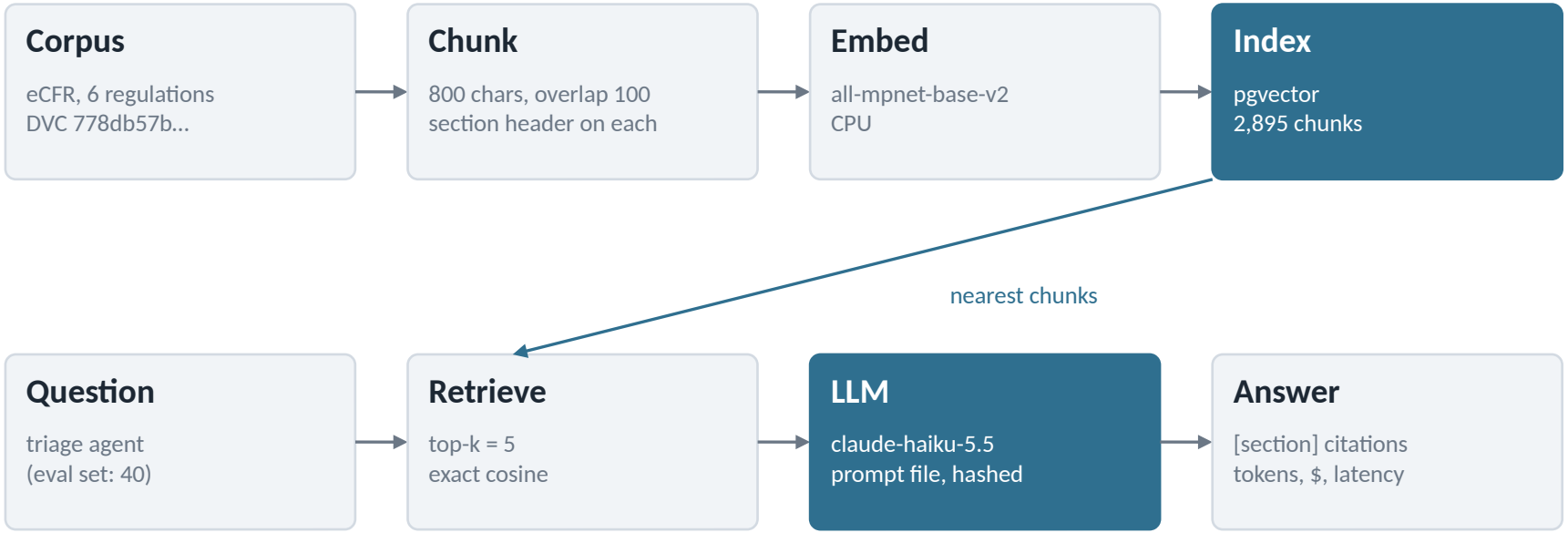
	LLM v2	Lab 2 model
Accuracy	0.880	0.895
Cost, 5,212 calls	$0.461	0
Latency p50 / p95	0.78 / 1.06 s	ms
Prompt hash	`cd3105724a2f`	—

But better on small classes

- Payday loans: F1 0.62 vs 0.46
- Debt management: 0.17 vs 0.00
- Money transfer: 0.79 vs 0.71

Corpus in, answer out, every setting in one file

`make rag-index`



`configs/llm.yaml: chunk_size, overlap, embedding_model, top_k, model, prompts`

Prompts and settings are versioned files

```yaml
# configs/llm.yaml
llm:
  model: anthropic/claude-haiku-5.5
  temperature: 0.0
  reasoning: false
index:
  chunk_size: 800
  chunk_overlap: 100
  embedding_model: all-mpnet-base-v2
retrieve:
  top_k: 5
answer:
  prompt: prompts/rag_answer.md
```

Prompt file	Hash on the run
classify.md v1	`88711daaddb5`
classify.md v2	`cd3105724a2f`
rag_answer.md	`answer_prompt_hash`
judge.md	`judge_prompt_hash`

Edit the file → new hash → new run

The old run stays reproducible from its commit. --set key=value changes one setting for one run, and the run logs it.

40 questions, a judge, and a check on the judge

40

questions with section ids and reference answers

14

rewritten by hand: quoted terms, stated thresholds

3

model families: Gemini writes, Claude answers, GPT judges

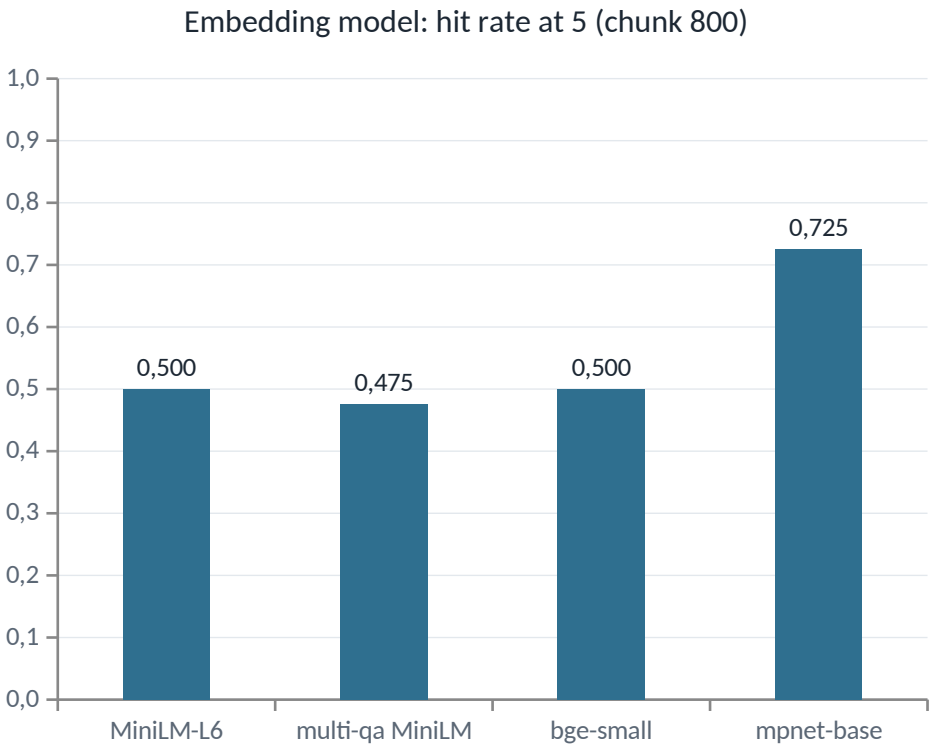
Metrics on every run

- Hit rate at 1, 3, 5, 10 and MRR@10
- Judge score 1–5 against the reference; pass = 4 or 5
- Citation valid: every cited section was retrieved
- Refusal rate

Is the judge right?

Check	Result
Answer judged against another question	0 / 40 pass
12 answers graded by hand	12 / 12 agree
Citation valid rate	1.00

The embedding model was the bottleneck, not chunking

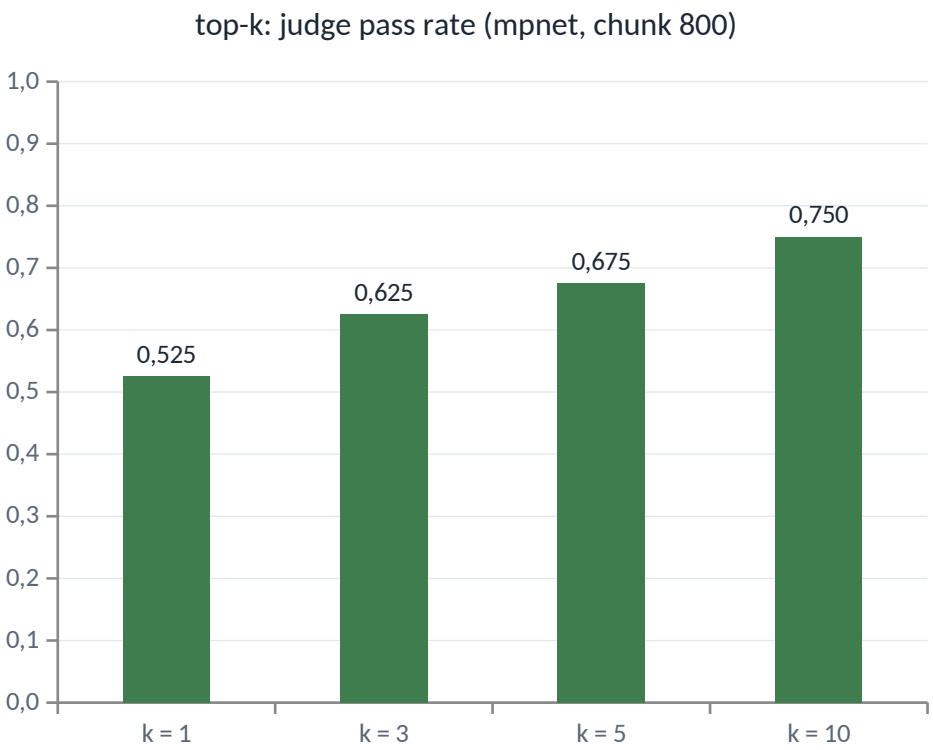


chunk_size	Hit@1	Hit@5	MRR
400	0.350	0.525	0.428
800	0.325	0.500	0.400
1600	0.275	0.575	0.399

mpnet: +9 questions of 40

Chunk size moves hit@5 by 1–3 questions. mpnet lifts MRR 0.40 → 0.57 and judge pass 0.43 → 0.65, and became the configured default.

More context answers more, at proportional cost



top_k	Tokens in	Cost (40 q)	p95
1	17,648	$0.0036	1.61 s
3	36,494	$0.0060	1.61 s
5	**55,709**	**$0.0080**	**1.88 s**
10	103,231	$0.0132	2.01 s

Noise floor: 1 question

The same config run twice gave pass 0.650 and 0.675. k = 5 stays configured; Lab 4 decides with the 10 s p95 budget.

An API, not local vLLM: what it costs either way

$0.0002
per regulation answer

$0.00009
per classified complaint

1.88 s
answer p95 (budget 10 s)

56 ms
retrieval p50, CPU

	API (chosen)	vLLM on this laptop
Money	per token, ≈ $13/month in the cloud estimate	free; a GPU ≈ $730/month
Latency	p95 1.9 s	1–2 s, 0.5B model only
Privacy	text leaves the machine	stays local
Memory	none	≈ 7 GB, swap already full
Reproducibility	model can change behind the name	weights pinned

Hidden cost found: default reasoning ate max_tokens, 20 of 40 answers empty. Now reasoning: false, and reasoning tokens are logged.

Lab 3 rows of the maturity scorecard

Property	Evidence
LLM component evaluated	40 questions; hit rate, MRR, judge as runs; baseline next to Lab 2
Prompts versioned	prompts/*.md; prompt and config hash on every run
Cost and latency recorded	tokens in/out/reasoning, $ as billed, p50/p95 per run
Index rebuilt by a third party	CI: same index_version 852a5450ef92, same hit@5 0.725

Run it

```
make corpus-public
make rag-index
make llm-baseline
make rag-eval
make rag-sweep
```

Needs OPENROUTER_API_KEY in .env; the index and retrieval need no key